



ADVANCES IN CARDIOLOGY AND THEIR IMPACT ON CI DEFINITIONS AND CLAIMS

Health Claims Forum; 26 February 2015

Dr. Andreas Armuss

1. The Cardiac CI Definition Conundrum
2. Aortic Valve: TAVI vs Open Surgery
3. Aorta: EVAR vs Graft Surgery
4. Heart Transplant: Human vs Artificial
5. Other Cardiac CI Conditions covered in UK
6. Questions we need to be asking ourselves

TAVI - Transcatheter Aortic Valve Implantation

The Cardiac Surgeon's Nightmare



Open heart surgery

Surgical valve replacement was only therapeutic option for the last 40 years

In 30–60% of elderly patients the risks of surgery outweigh the benefits

Transcatheter Aortic Valve Implantation (TAVI)

2002: first animal experiments

2006: first clinical results in (Germany)

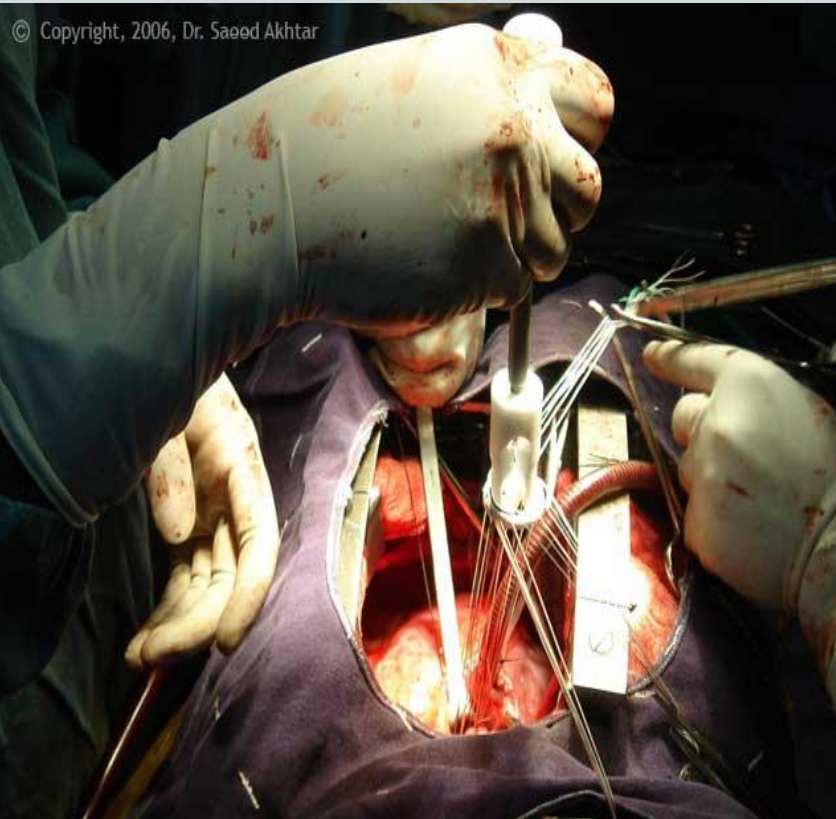
2010: TAVI vs medical therapy in **inoperable** patients

2011: TAVI vs open surgery in **high-risk** patients

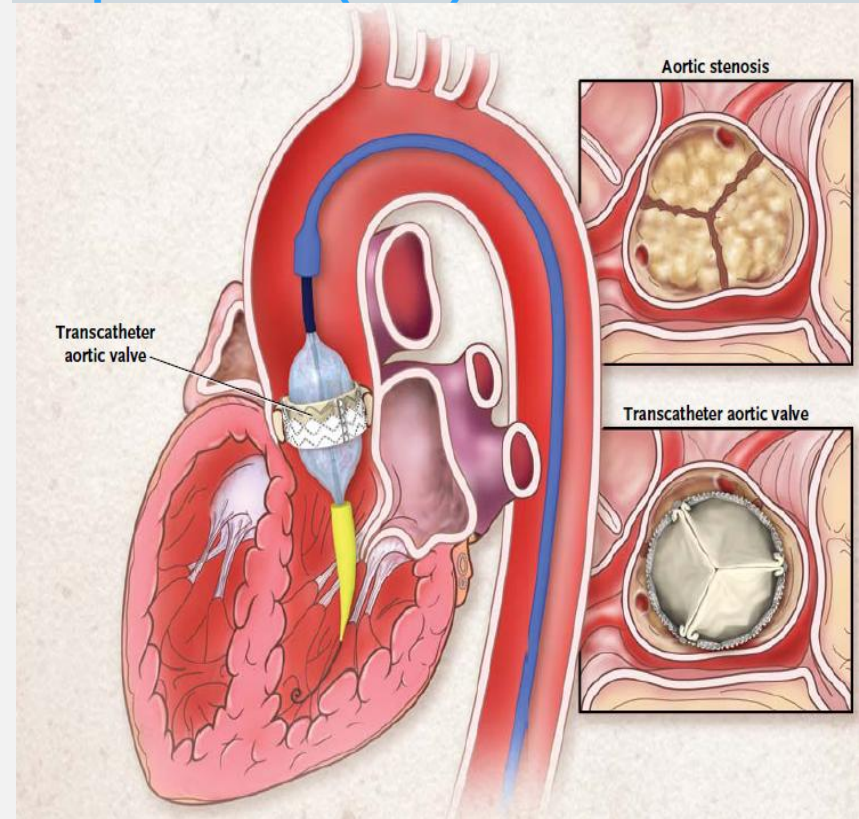
2012: TAVI (self-expanding bioprosthesis) vs medical therapy in **inoperable** patients

2014: TAVI (self-expanding bioprosthesis) vs open surgery in patients **at increased risk**

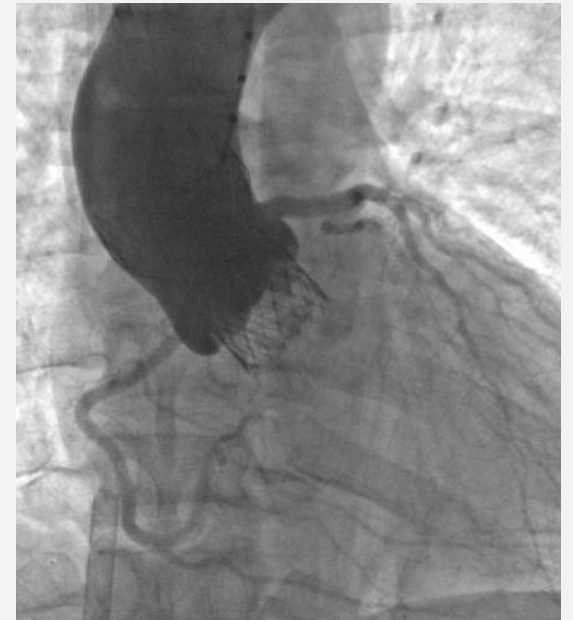
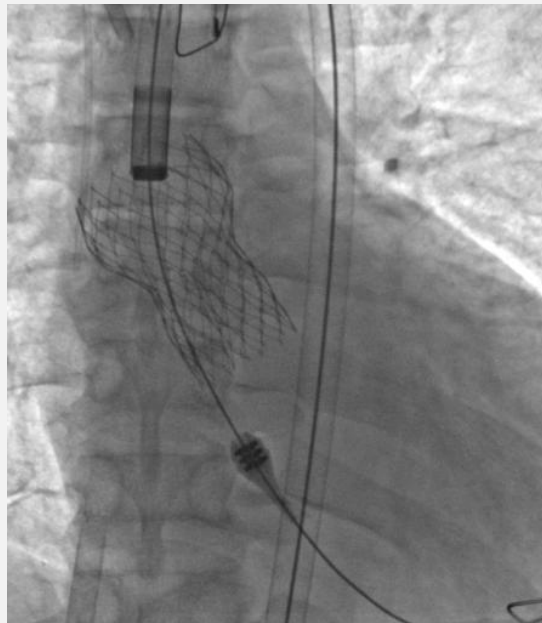
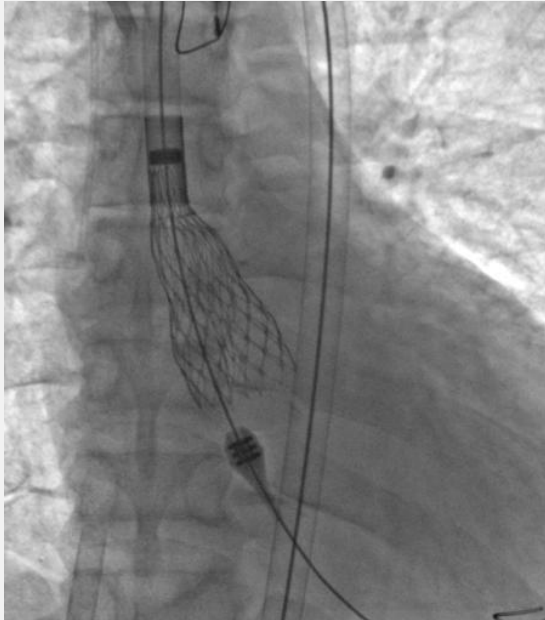
Open heart surgery



Transcatheter Aortic Valve Implantation (TAVI)

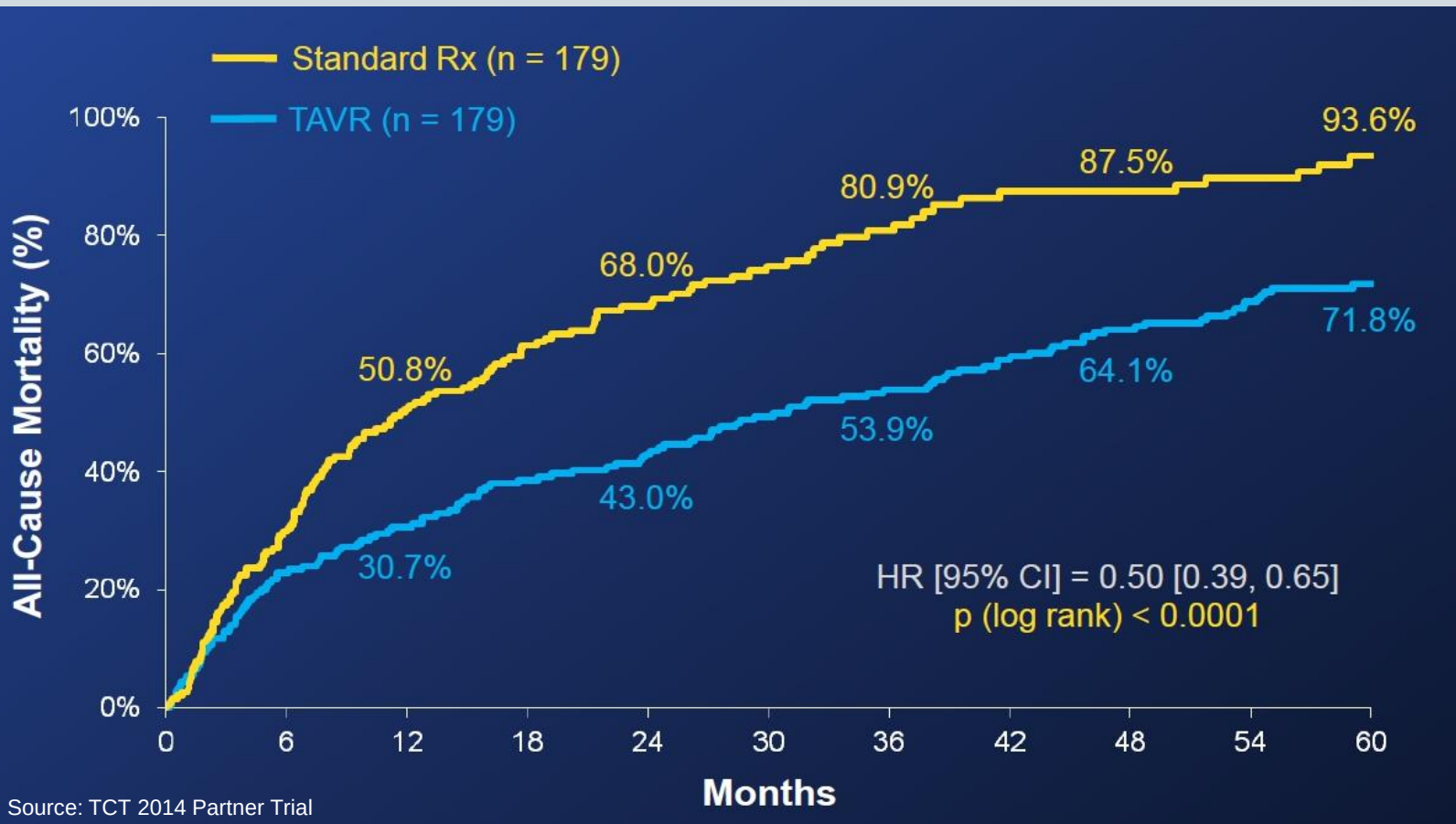


Transcatheter Aortic Valve Implantation (TAVI)



5 Year Results of PARTNER Trial - Inoperable Patients -

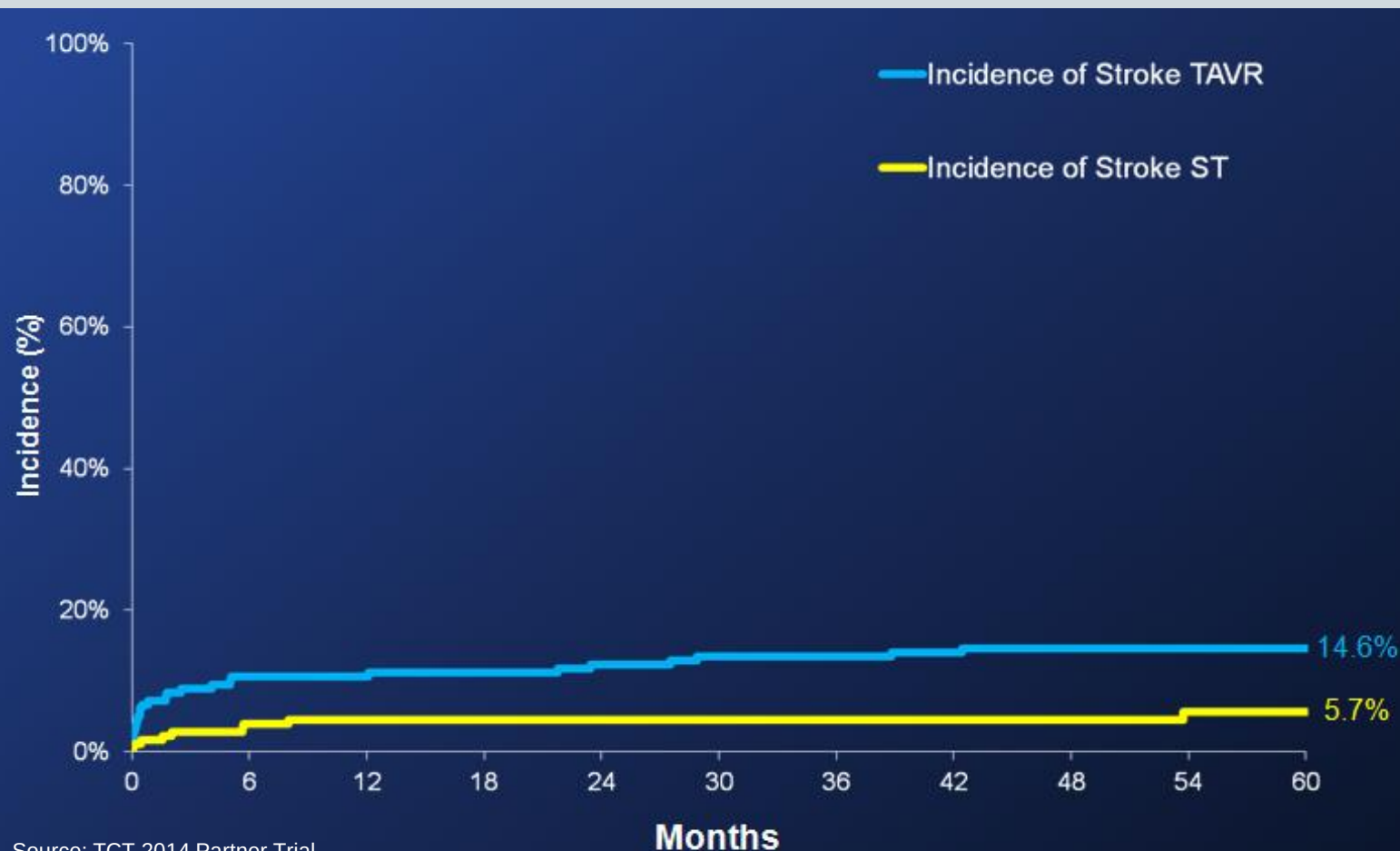
TAVI vs Standard (Medical) Treatment ; All Cause Mortality



Mean Age 83 years. US population without comorbidities: mortality at 5 years is 40.5%.

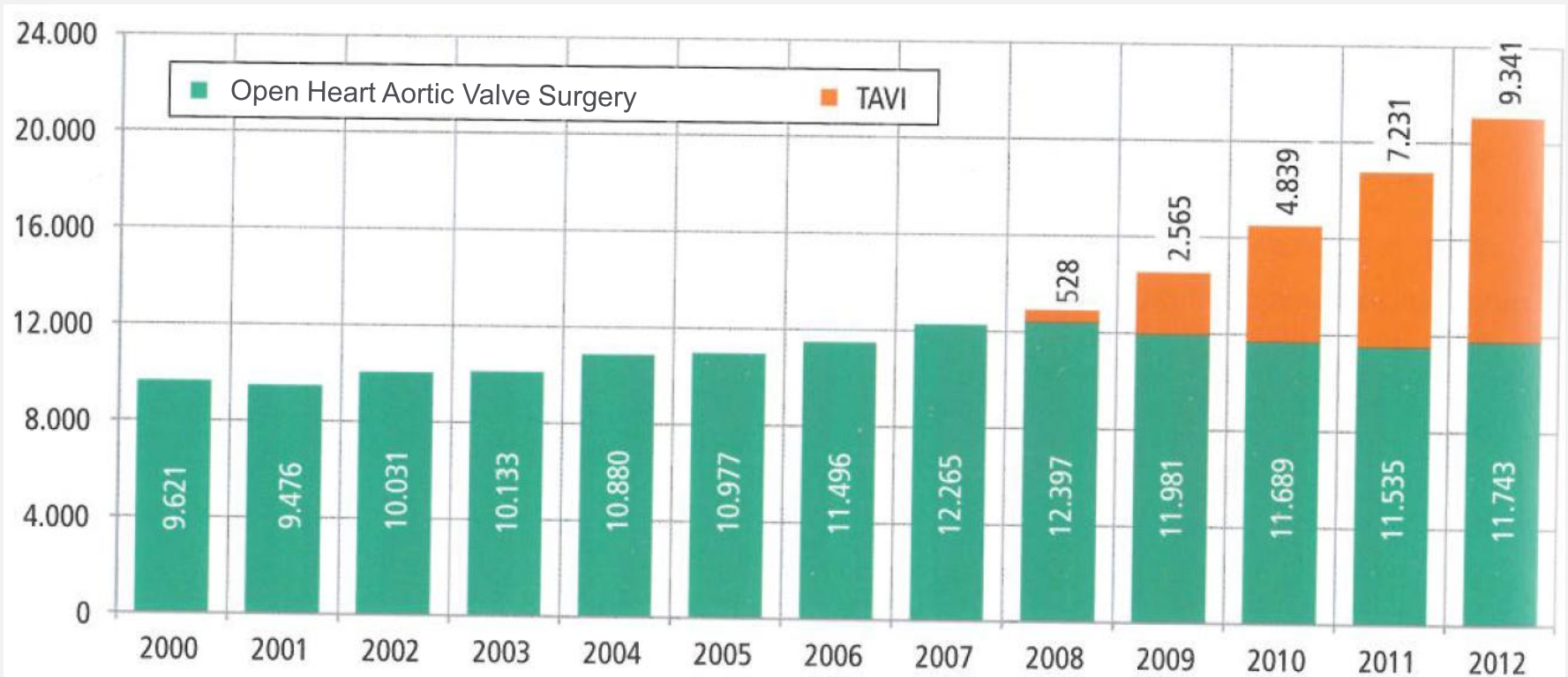
5 Year Results of PARTNER Trial - Inoperable Patients -

TAVI vs Standard (Medical) Treatment ; Incidence of Stroke



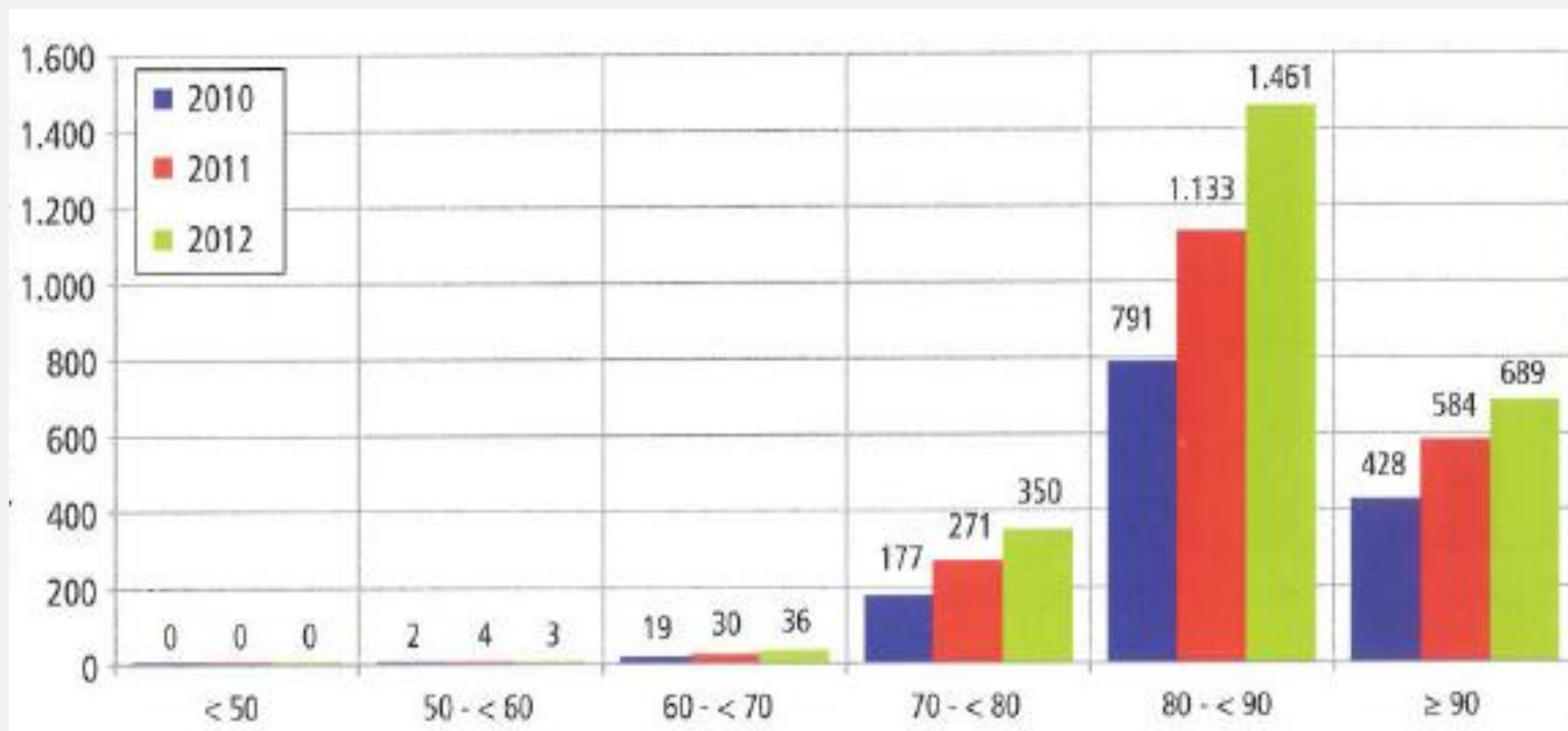
Source: TCT 2014 Partner Trial

Open Aortic Valve Surgery and TAVI Procedures in Germany 2000 - 2012



Source: Deutscher Herzbericht 2013

TAVI Procedures per 1 Million Population in Germany 2010 - 2012

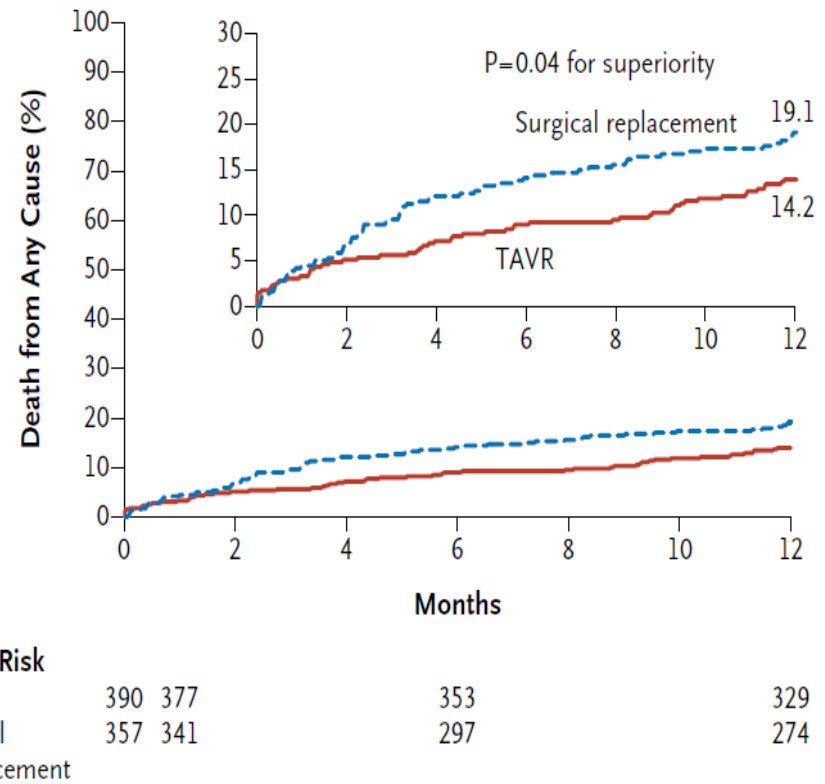


Source: Deutscher Herzbericht 2013

NEJM 2014: Transcatheter Aortic-Valve Replacement with a Self-Expanding Prosthesis

Table 1. Characteristics of the Patients at Baseline.*

Characteristic	Intention-to-Treat Population	
	TAVR Group (N=394)	Surgical Group (N=401)
Age — yr	83.2±7.1	83.5±6.3
Female sex — no. (%)	183 (46.4)	189 (47.1)
NYHA class — no. (%)		
Class II	56 (14.2)	53 (13.2)
Class III	258 (65.5)	277 (69.1)
Class IV	80 (20.3)	71 (17.7)
STS PROM estimate†		
Mean estimate — %	7.3±3.0	7.5±3.2
<4% — no. (%)	33 (8.4)	42 (10.5)
4–10% — no. (%)	308 (78.2)	288 (71.8)
>10% — no. (%)	53 (13.5)	71 (17.7)
Logistic EuroSCORE — %‡	17.6±13.0	18.4±12.8



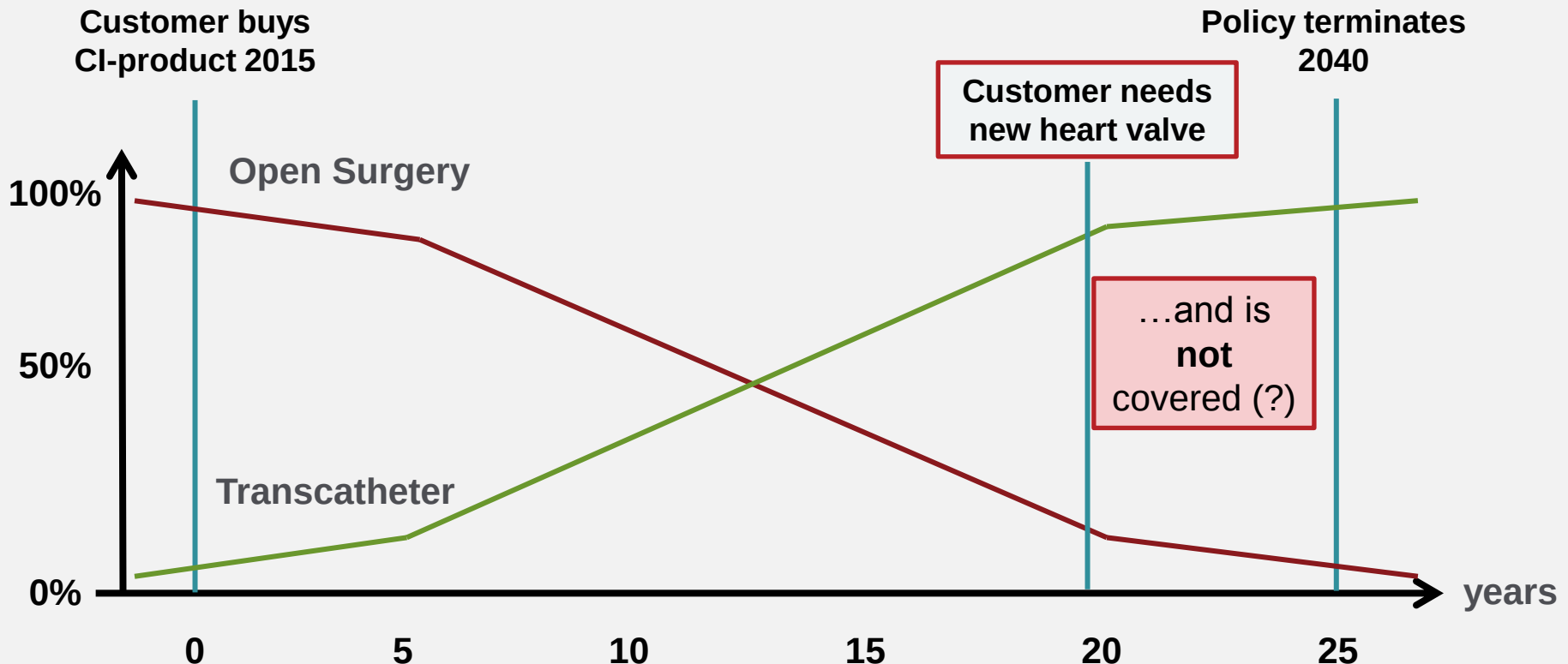
Risk of Stroke after 1 year: TAVI 8.8%; Open Surgery 12.6% !!

- Heart Valve Replacement or Repair -

Heart valve replacement or repair – *with surgery to divide the breastbone*

The undergoing of surgery requiring median sternotomy (surgery to divide the breastbone) on the advice of a Consultant Cardiologist to replace or repair one or more heart valves.

UK ABI Definition - Future Proof ? -



Insurance companies will (need to) pay out „transcatheter“ claims sooner or later

- TAVI already increases the number of aortic valve procedures in patients > 70 years
- New TAVI devices: reduced mortality and less complications if compared with open surgery even in patients with lower risk scores
- Future projection:
 - TAVI may become the standard procedure also in younger and healthier patients and numbers will increase (even more) significantly
 - However, in younger patients TAVI will replace open surgery without changing the indication for aortic valve replacement itself = no change in incidence rates
- To become future proof, CI definitions should include TAVI as a claims trigger
- It should be considered to pay out TAVI claims in claimants < 65 years of age.
- In claimants > 65 years of age, the decision to pay the claim should include the indication for TAVI (e.g. was the claimant eligible for open surgery or not).

Heart Valve Surgery – *including minimally invasive and catheter-based procedures*

The undergoing of surgery to replace or repair one or more defective heart valves. The following procedures are covered under this definition:

- Heart valve replacement or repair with full sternotomy (vertical division of the breastbone), partial sternotomy or thoracotomy
- Ross-Procedure
- Catheter-based valvuloplasty
- Transcatheter aortic valve implantation (TAVI)

The surgery must be determined to be medically necessary by a Consultant Cardiologist or Cardiac Surgeon and supported by echocardiogram or cardiac catheterisation findings.

For the above definition, the following are not covered:

- Transcatheter mitral valve clipping

TAVI

You can still
operate cardiac
myxomas.

OPEN SURGERY

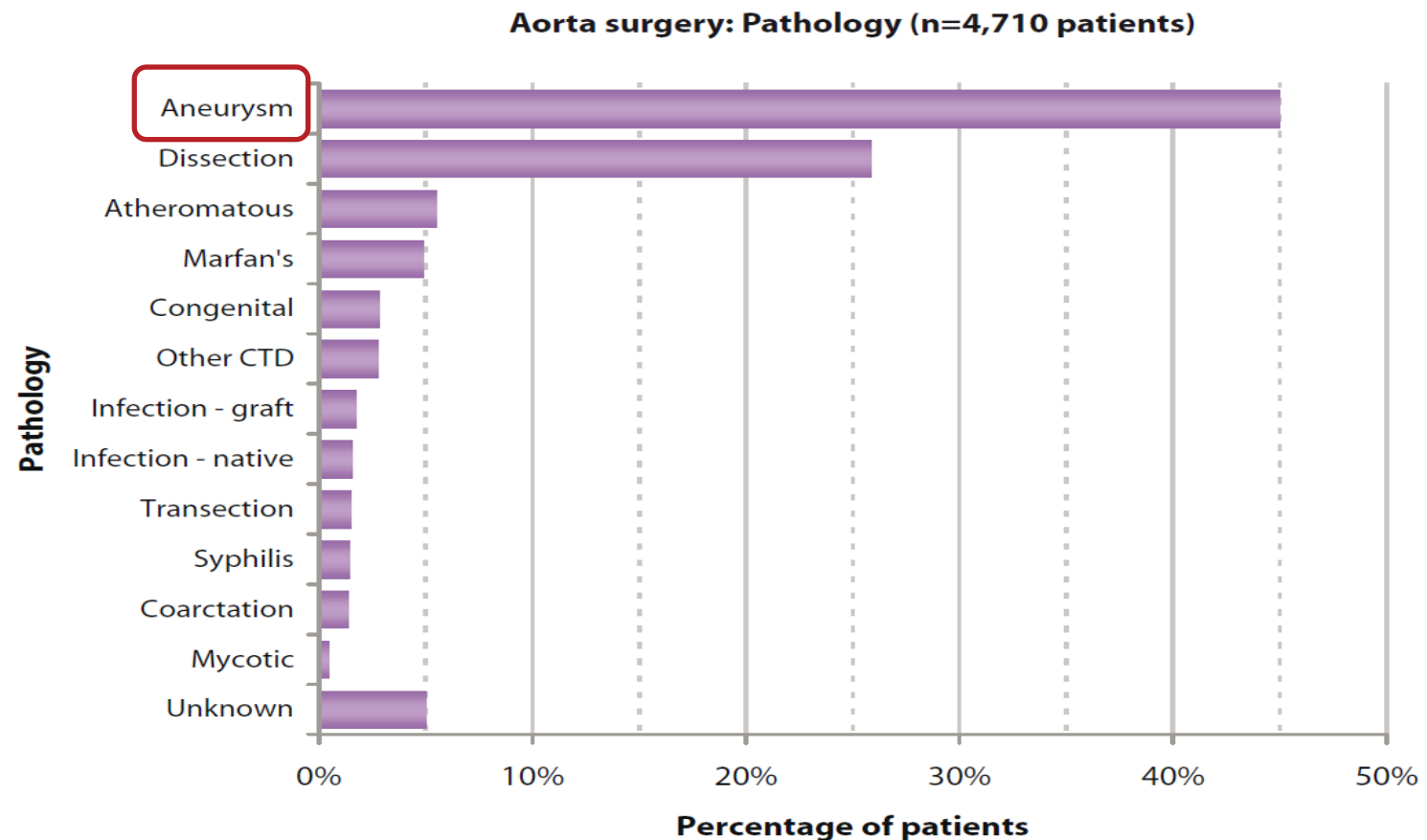
These bloody
cardiologists



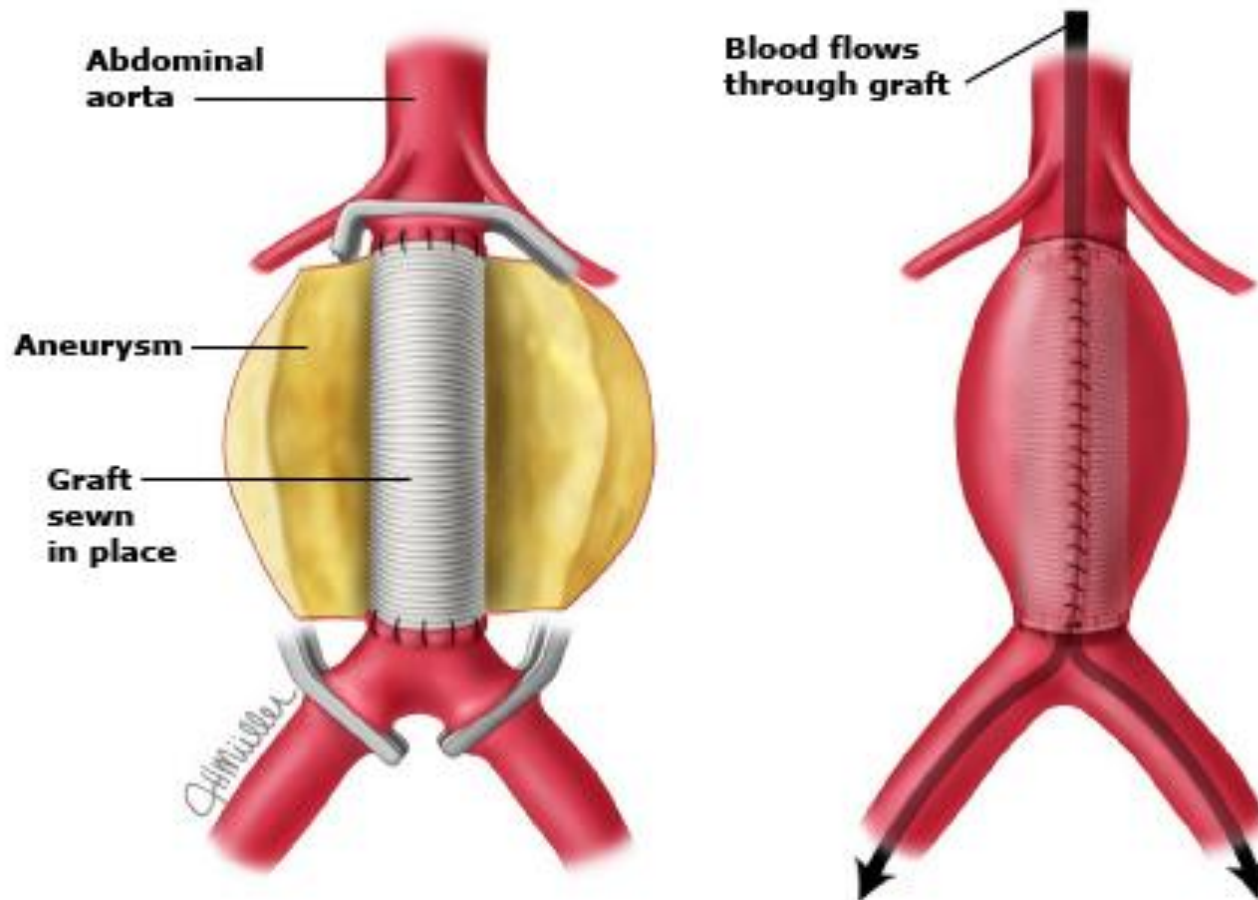
Endovascular Aneurysm Repair



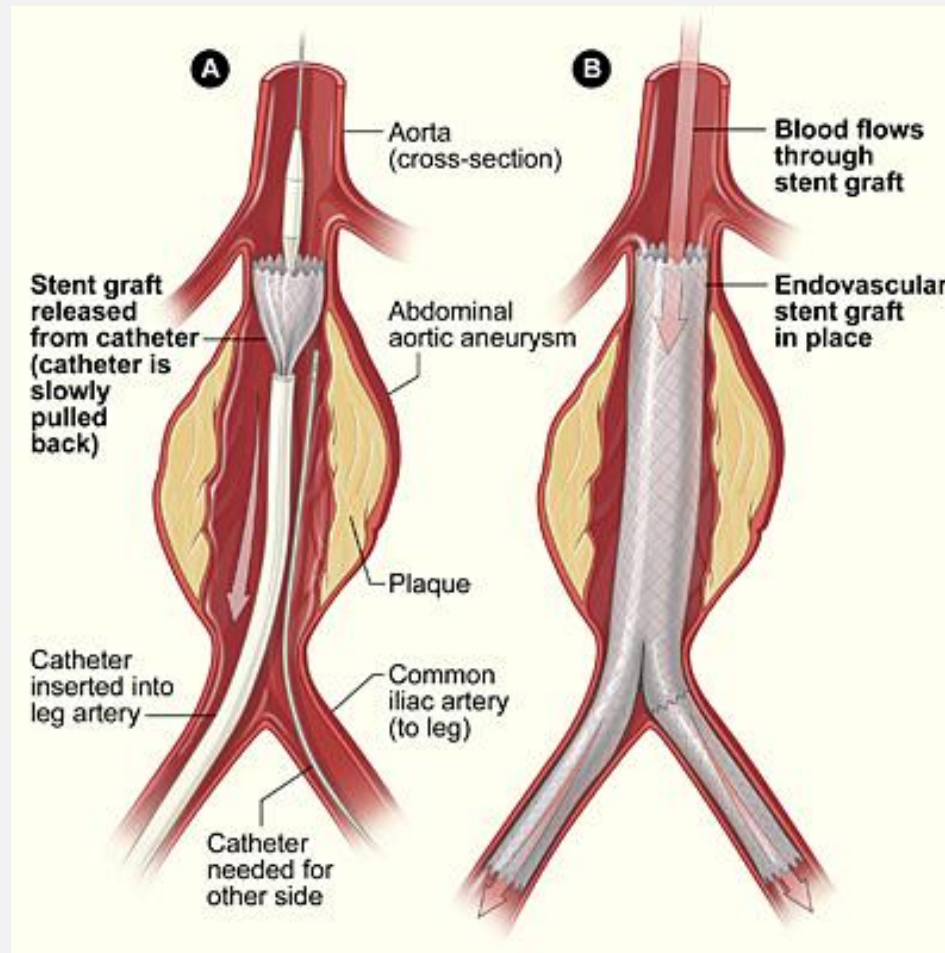
Sixth National Adult Cardiac Surgical Database Report 2008 of the Society for Cardiothoracic Surgery in Great Britain & Ireland



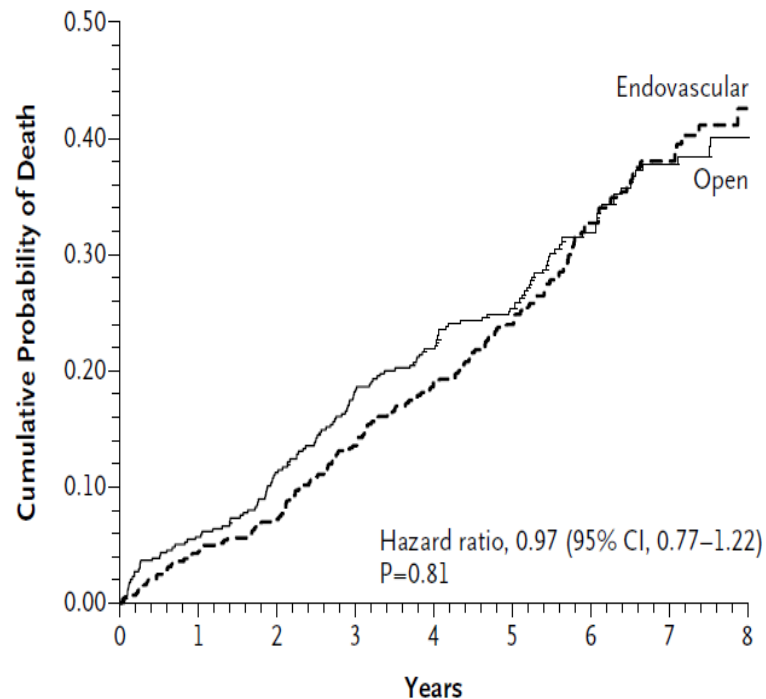
Open surgery with a graft:



Endovascular Aneurysm Repair (EVAR):



Mortality



No. at Risk

	0	1	2	3	4	5	6	7	8
Open	437	410	386	354	329	266	169	102	35
Endovascular	444	423	410	381	347	265	159	94	34

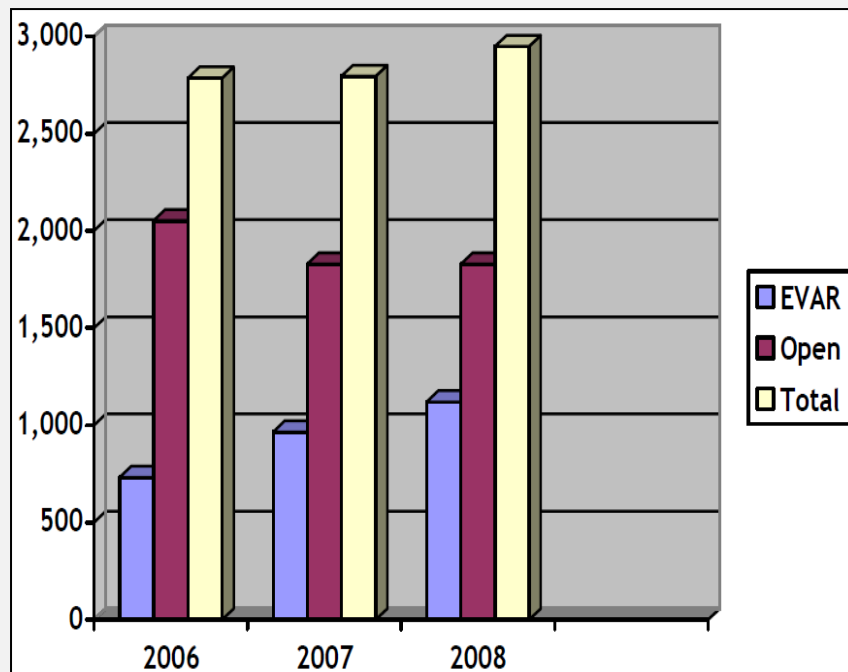
Morbidity

Frequencies of complications following EVAR and OAR

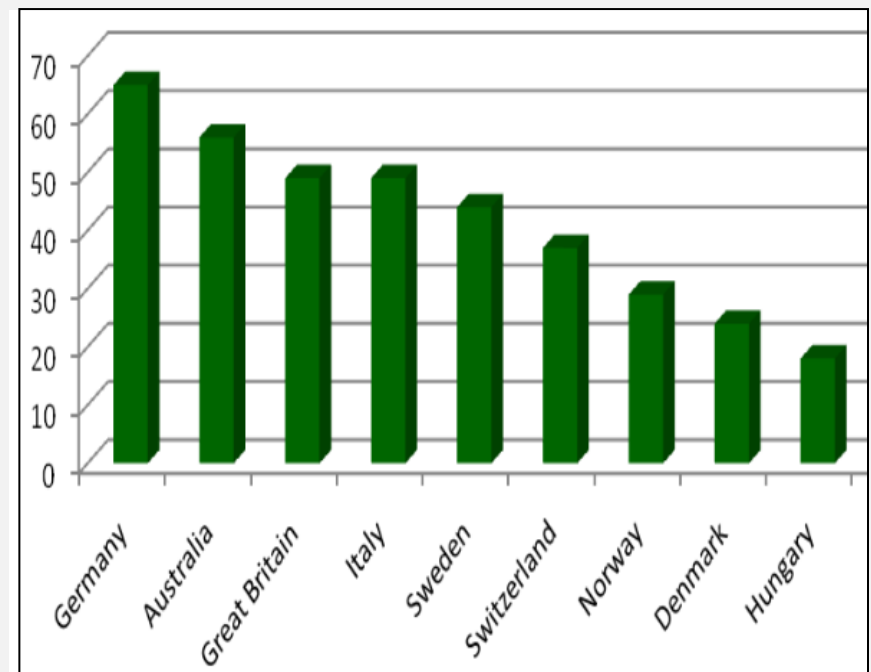
		OAR	EVAR
Early complications	Secondary bleeding	3%	<2%
	– Peripheral thromboembolic complications	1%	1%
	– Renal complications	1%	1 to 2%
	Colorectal ischemia	1%	1%
	Damage to the ureter/pelvic veins	<1%	–
	Erectile dysfunction	>60%	Significantly fewer
	Compromised wound healing	3%	3% (inguinal)
Late complications	Prosthesis infections	1 to 2%	<1%
	Closure of bypass leg	3%	6%
	Suture aneurysms	1%	–
	Incisional hernias	16%	–
	Prosthesis migration	–	5%
	Secondary aneurysm rupture	–	0.8%

Source: Guidelines of the German Society for Vascular Surgery and Vascular Medicine (Deutsche Gesellschaft für Gefäßchirurgie und Gefäßmedizin) (3)
OAR: open aortic repair; EVAR: endovascular aneurysm repair

**Total EVARs and Open Surgeries
Canada, 2006 to 2008**



**Rate of EVAR for AAA in different
countries during 2005-2009**



UK ABI Definition:

Aorta graft surgery – *for disease*

The undergoing of surgery for disease to the aorta with excision and surgical replacement of a portion of the diseased aorta with a graft. The term aorta includes the thoracic and abdominal aorta but not its branches.

For the above definition, the following are not covered:

- Any other surgical procedure, for example the insertion of stents or endovascular repair.
- Surgery following traumatic injury to the aorta.

- The actual ABI Definition excludes ~ 50% of all operations for abdominal aneurysm repair
- Rejecting an EVAR claim could be a reputational risk, because:
 - EVAR is mentioned in the NICE guidelines of the NHS
 - Mortality and Morbidity are comparable between EVAR and open surgery
 - There is no significant cost benefit of EVAR compared to open surgery
- As EVAR is (in most cases) replacing open surgery without widening the indication for the procedure, the impact on pricing seems to be limited.

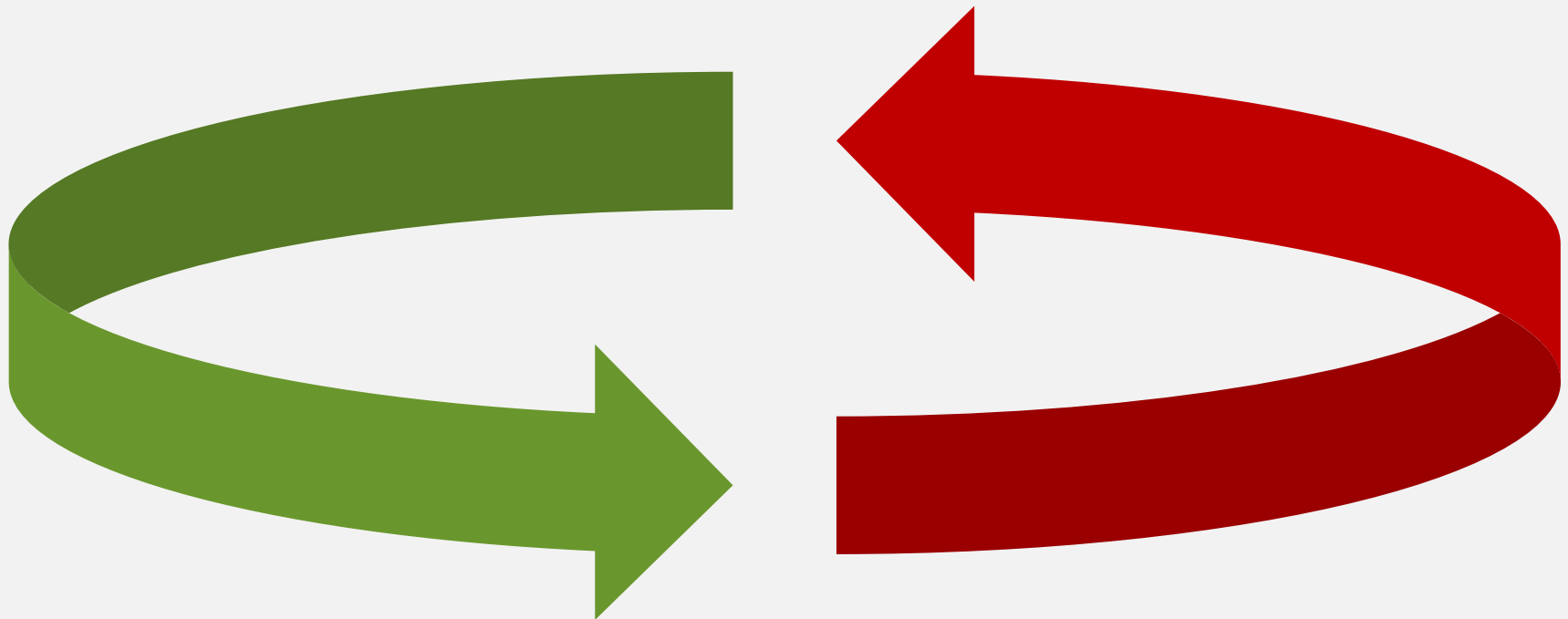
Surgery of the Aorta - *including minimally invasive procedures*

The undergoing of surgery to treat narrowing, obstruction, aneurysm or dissection of the aorta. Minimally invasive procedures like endovascular repair are covered under this definition. The surgery must be determined to be medically necessary by a Consultant Surgeon and supported by imaging findings.

For the above definition, the following are not covered:

- Surgery to any branches of the thoracic or abdominal aorta (including aortofemoral or aortoiliac bypass grafts)
- Surgery of the aorta related to hereditary connective tissue disorders (e.g. Marfan syndrome, Ehlers–Danlos syndrome)
- Surgery following traumatic injury to the aorta.

Heart Transplantation



Artificial Hearts

2 August 2011 Last updated at 19:23 GMT

4,140  Share    

Plastic heart gives dad Matthew Green new lease of life

A 40-year-old father who was dying from heart failure is set to leave hospital after receiving an artificial heart.

Matthew Green is ready to go home and await a transplant after surgeons at Papworth Hospital in Cambridgeshire replaced his heart with an implant.

His new plastic heart is powered by a portable driver in a backpack, which he said had "revolutionised" his life.

It is thought to be the first time a UK patient has been able to go home with an entirely artificial heart.

Around 900 similar operations have been carried out around the world.



Mr Green says he is excited about playing in the garden with his son

- Still shortage of donor hearts in most countries

Solution: **artificial hearts**

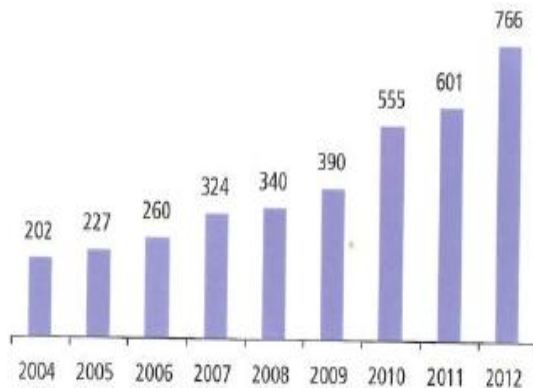
- First experiments already in the 1960s
- **Usually bridging to HTX**
- Since the 1990s artificial hearts are implanted into humans

Heart Transplantation

- Assist Devices and Total Artificial Hearts -

Assist Devices and Total Artificial Hearts Implanted in Germany 2004-2012

L/RVAD Implantation

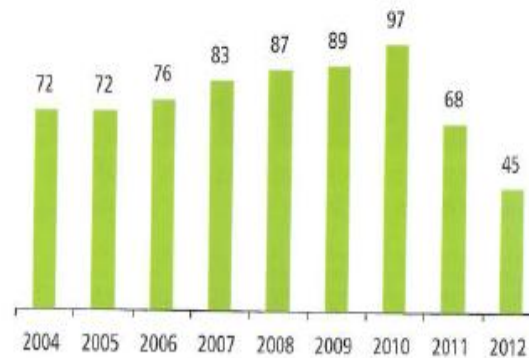


L/RVAD:

Left ventricular assist device

Right ventricular assist device

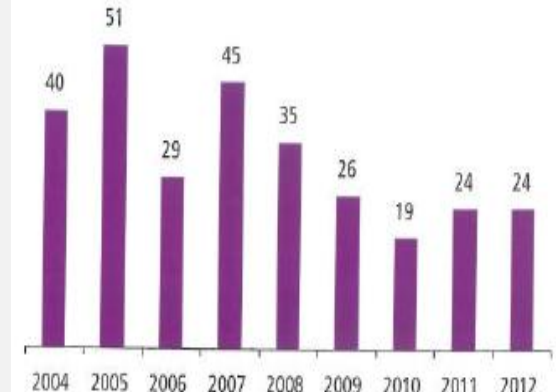
BVAD



BVAD:

Biventricular assist device

TAH



TAH:

Total artificial heart

UK ABI Definition

- Major Organ Transplant -

UK ABI Definition:

Major organ transplant – *from another person*

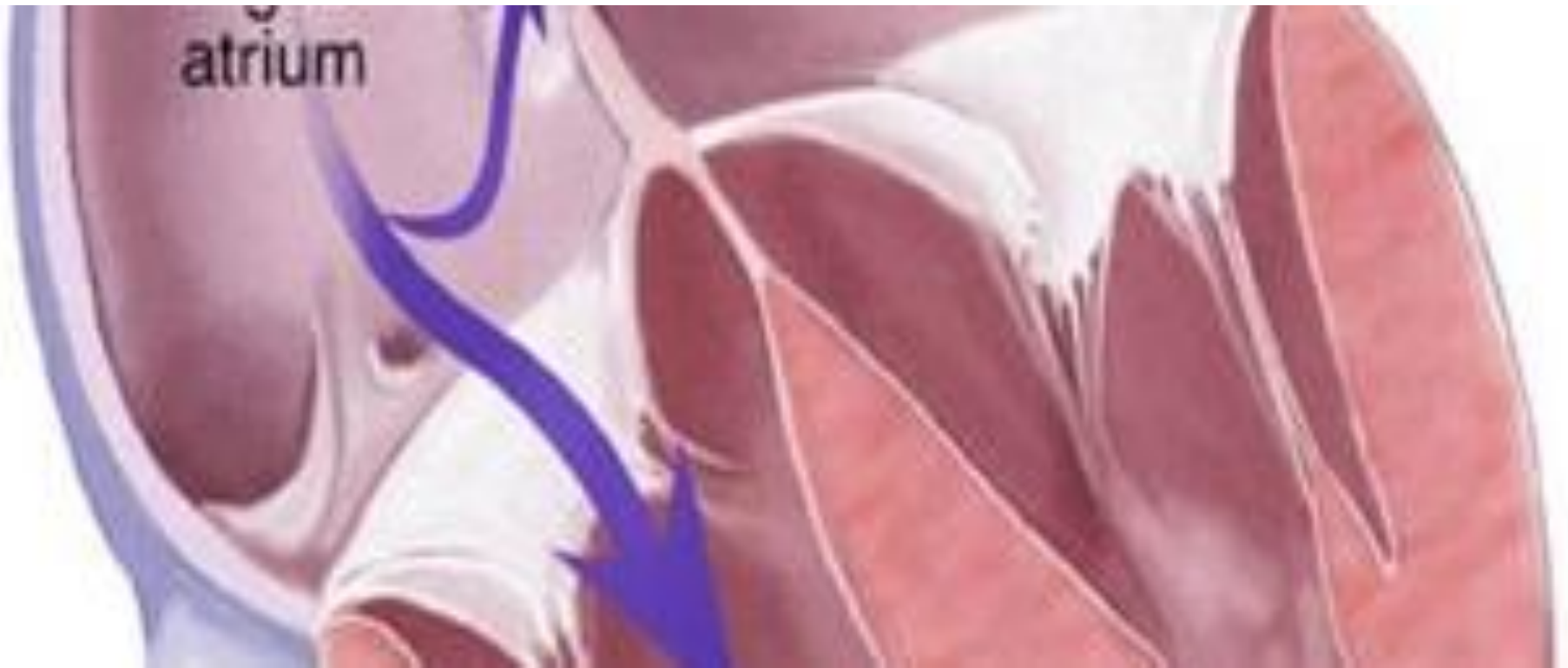
The undergoing as a recipient of a transplant of bone marrow or of a complete heart, kidney, liver, lung, or pancreas, or inclusion on an official UK waiting list for such a procedure.

For the above definition, the following is not covered:

- Transplant of any other organs, parts of organs, tissues or cells.

- The actual ABI Definition excludes all artificial hearts, as the new organ has to come from “another human”
- However, the inclusion on an official UK waiting list for such a procedure (!) is covered.
- It is difficult to say, if someone receiving an artificial heart is a valid claim under the ABI definition, because:
 - the organ is not coming from a human, but
 - most of the recipients are most likely included on a transplant list at some time
- The number of claimants with a TAH will remain small in the foreseeable future

Other Cardiac CI Conditions commonly covered in UK



Definition

- Coronary artery by-pass grafts -

UK ABI Definition:

Coronary artery by-pass grafts – *with surgery to divide the breastbone*

The undergoing of surgery requiring median **sternotomy** (surgery to divide the breastbone) on the advice of a Consultant Cardiologist to correct narrowing or blockage of one or more coronary arteries with by-pass grafts.

Minimally invasive direct coronary artery bypass (MIDCAB) has been common for over 10 years and approx. 5 to max. 10% of all Bypass-Surgeries will be MIDCAB

- The ABI Definition excludes MIDCAB
- Potential for reputational damage

UK Non-ABI Definition:

Cardiomyopathy

A definite diagnosis of cardiomyopathy by a Consultant Cardiologist. There must be clinical impairment of heart function resulting in the permanent loss of ability to perform physical activities to at least Class 3 of the New York Heart Association classifications of functional capacity (i.e. heart disease resulting in marked limitation of physical activities where less than ordinary activity causes fatigue, palpitation, breathlessness or chest pain).

For the above definition the following are not covered:

Cardiomyopathy secondary to alcohol or drug abuse.

All other forms of heart disease, heart enlargement and myocarditis.

How objective is NYHA classifications? Based on patient subjective reporting + potential for co-morbidities such as overweight, lung disease to impact function

Does the claims trigger make sense for all cardiomyopathies ?

Definition

- Angioplasty -

UK Non-ABI Definition:

Coronary Angioplasty

The undergoing of balloon angioplasty on the advice of a Consultant Cardiologist to correct narrowing or blockage of at least 70% in each of two or more coronary arteries.

What are “two coronary arteries” ? Are a main branch and a side branch already “two” ?

Potential for reputational damage – what about other procedures atherectomy, rotablation

Measuring diameter of narrowing subjective +/- 10% on standard angiogram – look to FFR (fractional flow reserve)

UK Non-ABI Definition:

Primary Pulmonary Hypertension

Primary Pulmonary hypertension with clinical impairment of heart function resulting in marked limitation of physical activities to at least Class 3 of the New York Heart Association's classification of functional capacity.

- Pulmonary Hypertension: 90% are secondary ! Needs to be clearly ruled out !
- How objective is NYHA classifications? Based on patient subjective reporting + potential for co-morbidities such as overweight, cardiac disease to impact function
- Definition not based on the guidelines for diagnosis of PPH

Questions we need to be asking ourselves

1. Can the ABI definitions continue to keep pace with medical advances ?
2. Are ABI+ definitions merely reflective of medical advances and common sense?
3. How can we as an industry effectively manage claims under older policy definitions?



THANK YOU VERY MUCH FOR YOUR KIND ATTENTION

Dr. Andreas Armuss